

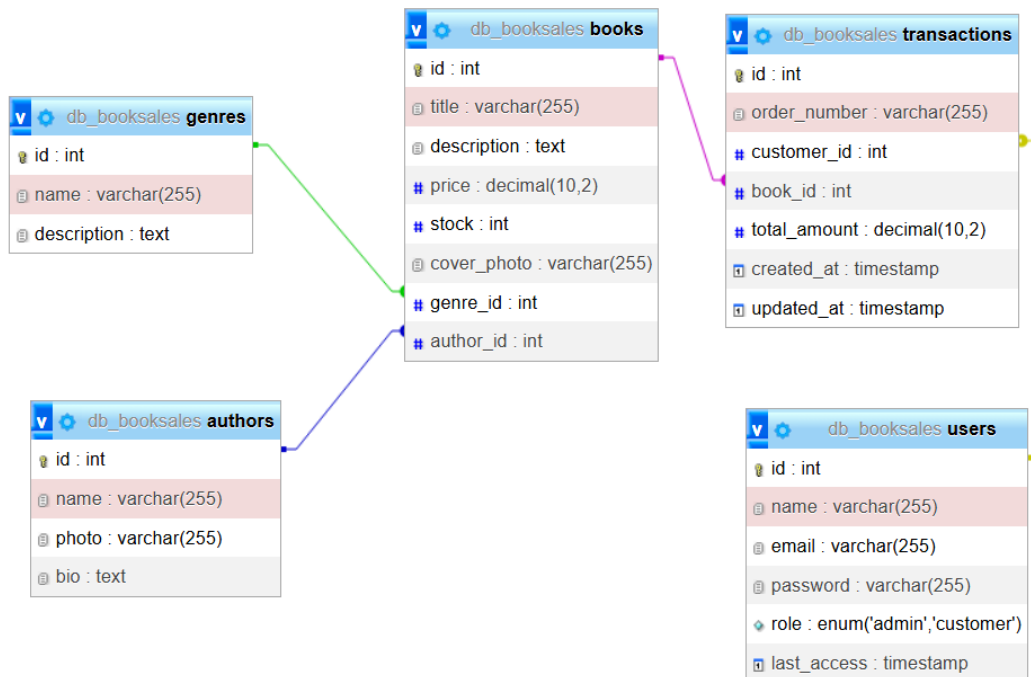
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TUGAS DATABASE 6

Link Repository

<https://github.com/ikhlaspp/SIB-NF-Academy-Batch-4-Full-Stack-Developer>

1. Desain database



2. Transaction 1

Pembelian Buku Sukses

Tabel

	id	order_number	customer_id	book_id	total_amount	created_at	updated_at
  Edit  Copy  Delete	1	ORD-001	1	1	75000.00	2026-04-11 09:19:48	2026-04-11 09:19:48
  Edit  Copy  Delete	2	ORD-002	2	3	150000.00	2026-04-11 09:19:48	2026-04-11 09:19:48
  Edit  Copy  Delete	3	ORD-003	4	2	85000.00	2026-04-11 09:19:48	2026-04-11 09:19:48
  Edit  Copy  Delete	4	ORD-004	5	5	60000.00	2026-04-11 09:19:48	2026-04-11 09:19:48
  Edit  Copy  Delete	5	ORD-005	1	4	120000.00	2026-04-11 09:19:48	2026-04-11 09:19:48

	id	title	description	price	stock	cover_photo	genre_id	author_id
  Edit  Copy  Delete	1	Laskar Pelangi	Kisah 10 anak di Belitung.	75000.00	20	laskar.jpg	1	1
  Edit  Copy  Delete	2	Bumi	Petualangan Raib di dunia paralel.	85000.00	15	bumi.jpg	2	2
  Edit  Copy  Delete	3	Harry Potter dan Batu Bertuah	Awal kisah penyihir cilik.	150000.00	10	hp1.jpg	2	3
  Edit  Copy  Delete	4	Bumi Manusia	Kisah cinta Minke dan Annelies.	120000.00	8	bumimanusia.jpg	5	4
  Edit  Copy  Delete	5	Kambing Jantan	Catatan harian pelajar bodoh.	60000.00	25	kambing.jpg	3	5

Query

```
1 START TRANSACTION;
2
3 INSERT INTO transactions (order_number, customer_id, book_id, total_amount)
4 VALUES ('ORD-006', 1, 1, 75000.00);
5
6 UPDATE books
7 SET stock = stock - 1
8 WHERE id = 1;
9
10 COMMIT;
```

Hasil Query (transaksi bertambah dan stok buku berkurang)

	id	order_number	customer_id	book_id	total_amount	created_at	updated_at
  Edit  Copy  Delete	1	ORD-001	1	1	75000.00	2026-04-11 09:19:48	2026-04-11 09:19:48
  Edit  Copy  Delete	2	ORD-002	2	3	150000.00	2026-04-11 09:19:48	2026-04-11 09:19:48
  Edit  Copy  Delete	3	ORD-003	4	2	85000.00	2026-04-11 09:19:48	2026-04-11 09:19:48
  Edit  Copy  Delete	4	ORD-004	5	5	60000.00	2026-04-11 09:19:48	2026-04-11 09:19:48
  Edit  Copy  Delete	6	ORD-006	1	1	75000.00	2026-04-16 11:24:54	2026-04-16 11:24:54

	id	title	description	price	stock	cover_photo	genre_id	author_id
  Edit  Copy  Delete	1	Laskar Pelangi	Kisah 10 anak di Belitung.	80000.00	29	laskar.jpg	1	1
  Edit  Copy  Delete	2	Bumi	Petualangan Raib di dunia paralel.	85000.00	15	bumi.jpg	2	2
  Edit  Copy  Delete	3	Harry Potter dan Batu Bertuah	Awal kisah penyihir cilik.	150000.00	10	hp1.jpg	2	3
  Edit  Copy  Delete	4	Bumi Manusia	Kisah cinta Minke dan Annelies.	120000.00	8	bumimanusia.jpg	5	4
  Edit  Copy  Delete	5	Kambing Jantan	Catatan harian pelajar bodoh.	60000.00	25	kambing.jpg	3	5

3. Transaction 2

Pembelian Gagal karena Sistem Error (Rollback)

Tabel

← →			▼	id	order_number	customer_id	book_id	total_amount	created_at	updated_at
	Edit	Copy	Delete	1	ORD-001		1	75000.00	2026-04-11 09:19:48	2026-04-11 09:19:48
	Edit	Copy	Delete	2	ORD-002		2	150000.00	2026-04-11 09:19:48	2026-04-11 09:19:48
	Edit	Copy	Delete	3	ORD-003		4	85000.00	2026-04-11 09:19:48	2026-04-11 09:19:48
	Edit	Copy	Delete	4	ORD-004		5	60000.00	2026-04-11 09:19:48	2026-04-11 09:19:48
	Edit	Copy	Delete	5	ORD-005		1	120000.00	2026-04-11 09:19:48	2026-04-11 09:19:48

Query

```
1 START TRANSACTION;
2
3 INSERT INTO transactions (order_number, customer_id, book_id, total_amount)
4 VALUES ('ORD-ERR-001', 999, 1, 75000.00);
5
6 ROLLBACK;
```

Hasil Query (Terjadi error dan tidak ada perubahan di tabel)

Error

SQL query: [Copy](#)

INSERT INTO transactions (order_number, customer_id, book_id, total_amount)
VALUES ('ORD-ERR-001', 999, 1, 75000.00);

MySQL said: 

#1452 - Cannot add or update a child row: a foreign key constraint fails ('db_booksales`.`transactions`, CONSTRAINT `transactions_ibfk_1` FOREIGN KEY ('customer_id')
REFERENCES `users` (`id`) ON DELETE CASCADE)

				id	order_number	customer_id	book_id	total_amount	created_at	updated_at
☐	 Edit	 Copy	 Delete	1	ORD-001		1	75000.00	2026-04-11 09:19:48	2026-04-11 09:19:48
☐	 Edit	 Copy	 Delete	2	ORD-002		2	150000.00	2026-04-11 09:19:48	2026-04-11 09:19:48
☐	 Edit	 Copy	 Delete	3	ORD-003		4	85000.00	2026-04-11 09:19:48	2026-04-11 09:19:48
☐	 Edit	 Copy	 Delete	4	ORD-004		5	60000.00	2026-04-11 09:19:48	2026-04-11 09:19:48
☐	 Edit	 Copy	 Delete	5	ORD-005		1	120000.00	2026-04-11 09:19:48	2026-04-11 09:19:48

4. Procedure 1

Prosedur untuk Update Stok Buku

Tabel

				id	title	description	price	stock	cover_photo	genre_id	author_id
☐	 Edit	 Copy	 Delete	1	Laskar Pelangi	Kisah 10 anak di Belitung.	75000.00	20	laskar.jpg	1	1
☐	 Edit	 Copy	 Delete	2	Bumi	Petualangan Raib di dunia paralel.	85000.00	15	bumi.jpg	2	2
☐	 Edit	 Copy	 Delete	3	Harry Potter dan Batu Bertuah	Awal kisah penyihir cilik.	150000.00	10	hp1.jpg	2	3
☐	 Edit	 Copy	 Delete	4	Bumi Manusia	Kisah cinta Minke dan Annelies.	120000.00	8	bumimanusia.jpg	5	4
☐	 Edit	 Copy	 Delete	5	Kambing Jantan	Catatan harian pelajar bodoh.	60000.00	25	kambing.jpg	3	5

Query

```
1 DELIMITER //
2
3 CREATE PROCEDURE TambahStokBuku (
4     IN p_book_id INT,
5     IN p_tambahan_stok INT
6 )
7 BEGIN
8     UPDATE books
9     SET stock = stock + p_tambahan_stok
10    WHERE id = p_book_id;
11 END //
12
13 DELIMITER ;
14
15 CALL TambahStokBuku(3, 10);
```

Hasil Query (Buku dengan id 3 stoknya bertambah 10)

⏪ ⏩		▼	id	title	description	price	stock	cover_photo	genre_id	author_id
☐	 Edit  Copy  Delete		1	Laskar Pelangi	Kisah 10 anak di Belitung.	80000.00	29	laskar.jpg	1	1
☐	 Edit  Copy  Delete		2	Bumi	Petualangan Raib di dunia paralel.	85000.00	15	bumi.jpg	2	2
☐	 Edit  Copy  Delete		3	Harry Potter dan Batu Bertuah	Awal kisah penyihir cilik.	150000.00	20	hp1.jpg	2	3
☐	 Edit  Copy  Delete		4	Bumi Manusia	Kisah cinta Minke dan Annelies.	120000.00	8	bumimanusia.jpg	5	4
☐	 Edit  Copy  Delete		5	Kambing Jantan	Catatan harian pelajar bodoh.	60000.00	25	kambing.jpg	3	5

5. Procedure 2

Prosedur Lengkap Proses Pembelian

Tabel

←T→ ▼	id	title	description	price	stock	cover_photo	genre_id	author_id
☐ Edit Copy Delete	1	Laskar Pelangi	Kisah 10 anak di Belitung.	75000.00	20	laskar.jpg	1	1
☐ Edit Copy Delete	2	Bumi	Petualangan Raib di dunia paralel.	85000.00	15	bumi.jpg	2	2
☐ Edit Copy Delete	3	Harry Potter dan Batu Bertuah	Awal kisah penyihir cilik.	150000.00	10	hp1.jpg	2	3
☐ Edit Copy Delete	4	Bumi Manusia	Kisah cinta Minke dan Annelies.	120000.00	8	bumimanusia.jpg	5	4
☐ Edit Copy Delete	5	Kambing Jantan	Catatan harian pelajar bodoh.	60000.00	25	kambing.jpg	3	5

←T→ ▼	id	order_number	customer_id	book_id	total_amount	created_at	updated_at
☐ Edit Copy Delete	1	ORD-001		1	75000.00	2026-04-11 09:19:48	2026-04-11 09:19:48
☐ Edit Copy Delete	2	ORD-002		3	150000.00	2026-04-11 09:19:48	2026-04-11 09:19:48
☐ Edit Copy Delete	3	ORD-003		2	85000.00	2026-04-11 09:19:48	2026-04-11 09:19:48
☐ Edit Copy Delete	4	ORD-004		5	60000.00	2026-04-11 09:19:48	2026-04-11 09:19:48
☐ Edit Copy Delete	5	ORD-005		4	120000.00	2026-04-11 09:19:48	2026-04-11 09:19:48

Query

```
1 DELIMITER //
2
3 CREATE PROCEDURE ProsesCheckoutAman (
4     IN p_order_number VARCHAR(255),
5     IN p_customer_id INT,
6     IN p_book_id INT
7 )
8 BEGIN
9     DECLARE v_harga DECIMAL(10,2);
10    DECLARE v_stok INT;
11
12    DECLARE EXIT HANDLER FOR SQLEXCEPTION
13    BEGIN
14        ROLLBACK;
15        SELECT 'GAGAL: Terjadi kesalahan sistem. Pastikan ID Pelanggan atau ID Buku valid!' AS Pesan;
16    END;
17
18    SELECT price, stock INTO v_harga, v_stok
19    FROM books
20    WHERE id = p_book_id;
21
22    IF v_stok > 0 THEN
23        START TRANSACTION;
24
25        INSERT INTO transactions (order_number, customer_id, book_id, total_amount)
26        VALUES (p_order_number, p_customer_id, p_book_id, v_harga);
27
28        UPDATE books
29        SET stock = stock - 1
30        WHERE id = p_book_id;
31
32        COMMIT;
33        SELECT 'SUKSES: Transaksi berhasil dan stok telah dikurangi.' AS Pesan;
34
35    ELSE
36        SELECT 'GAGAL: Stok buku sudah habis.' AS Pesan;
37    END IF;
38
39 END //
40
41 DELIMITER ;
42 CALL ProsesCheckoutAman('ORD-999', 4, 5);
```

Hasil Query (Query sukses, transaksi bertambah dan stok buku dikurangi)

✓ Showing rows 0 - 0 (1 total, Query took 0.0180 seconds.)

`CALL ProsesCheckoutAman('ORD-999', 4, 5);`

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]

☐ Show all | Number of rows: 25

Extra options

Pesan

SUKSES: Transaksi berhasil dan stok telah dikurang...

←T→		id	title	description	price	stock	cover_photo	genre_id	author_id
☐  Edit  Copy  Delete	1	Laskar Pelangi	Kisah 10 anak di Belitung.	80000.00	29	laskar.jpg	1	1	
☐  Edit  Copy  Delete	2	Bumi	Petualangan Raib di dunia paralel.	85000.00	15	bumi.jpg	2	2	
☐  Edit  Copy  Delete	3	Harry Potter dan Batu Bertuah	Awal kisah penyihir cilik.	150000.00	20	hp1.jpg	2	3	
☐  Edit  Copy  Delete	4	Bumi Manusia	Kisah cinta Minke dan Annelies.	120000.00	8	bumimanusia.jpg	5	4	
☐  Edit  Copy  Delete	5	Kambing Jantan	Catatan harian pelajar bodoh.	60000.00	24	kambing.jpg	3	5	

←T→		id	order_number	customer_id	book_id	total_amount	created_at	updated_at
☐  Edit  Copy  Delete	1	ORD-001		1	1	75000.00	2026-04-11 09:19:48	2026-04-11 09:19:48
☐  Edit  Copy  Delete	2	ORD-002		2	3	150000.00	2026-04-11 09:19:48	2026-04-11 09:19:48
☐  Edit  Copy  Delete	3	ORD-003		4	2	85000.00	2026-04-11 09:19:48	2026-04-11 09:19:48
☐  Edit  Copy  Delete	4	ORD-004		5	5	60000.00	2026-04-11 09:19:48	2026-04-11 09:19:48
☐  Edit  Copy  Delete	6	ORD-006		1	1	75000.00	2026-04-16 11:24:54	2026-04-16 11:24:54
☐  Edit  Copy  Delete	8	ORD-999		4	5	60000.00	2026-04-16 11:33:23	2026-04-16 11:33:23